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Domestic appliance, operating element for a domestic appliance, and method for controlling a domestic appliance

Abstract

A household appliance includes an operating region for controlling the household appliance, and an operating element configured for attachment on the operating region by magnetic force and removal therefrom and being movable relative to the operating region. The operating element includes an outer element and an inner element which is rotatable relative to the outer element. The inner element includes a user interface element and a magnetic body, by which the inner element can be aligned relative to the outer element by a predefined magnetic field.

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Background/Summary

CROSS-REFERENCES TO RELATED APPLICATIONS

(1) This application is the U.S. National Stage of International Application No. PCT/EP2017/080217, filed Nov. 23, 2017, which designated the United States and has been published as International Publication No. WO 2018/104066 A1 and which claims the priority of German Patent Application, Serial No. 10 2016 224 302.0, filed Dec. 7, 2016, pursuant to 35 U.S.C. 119 (a)-(d).

BACKGROUND OF THE INVENTION

(2) The present invention relates to a household appliance, an operating element for a household appliance and a method for controlling a household appliance with such an operating element.

(3) Known household appliances, such as cooking appliances for example, are operated by means of a touch-sensitive input surface, which is also equipped with display elements for example to display an operating mode of the household appliance. The advantage of such an operating element is for example that the surface of the cooking appliance is easy to clean. However, many users prefer haptic operating elements, in particular rotary knobs, for setting a cooking temperature or cooking time for example.

(4) DE 10 2012 213 693 A1 describes a two-part operating element, which can be positioned on such an input surface, its top being supported in a rotatable manner when in position. The operating element can be removed after use, thereby offering the advantages of the input surface described above as well as the advantages of a haptic operating element.

(5) DE 10 2014 223 371 A1 describes a similar system, in which the touch-sensitive input surface is configured as a touchscreen. The touchscreen adjusts the displayed graphic user interface as a function of the position of the operating element.

(6) Further conventional household appliances are known from WO 2010 143 244 A1, WO 2012 080 156 A1, DE 10 2011 007 471 A1, DE 10 2006 026 187 A1 and EP 225 17 62 A2.

BRIEF SUMMARY OF THE INVENTION

(7) Against this background, one object of the invention is to propose a household appliance with an improved operating element.

(8) According to a first aspect a household appliance with at least one operating region for controlling the household appliance and an operating element is proposed. The operating element can be held on the operating region by means of magnetic force and can be moved relative to the operating region. The operating element can also be positioned in such a manner that it can be removed from the operating region. The operating element has an inner element and an outer element. The inner element can be rotated relative to the outer element and has a user interface element and a magnetic body. The inner element can be aligned relative to the outer element by the magnetic body by means of a predefined magnetic field.

(9) Such a household appliance has the advantage, in particular, that a number of operating regions can be provided on the household appliance, being arranged differently for example, perhaps horizontally or vertically, and each able to be operated using the one operating element. The operating element here allows a haptic user experience. It also ensures that the user interface element of the inner element of the operating element is always correctly aligned by the predefined

magnetic field. This generally facilitates operation of the household appliance, reducing the probability of incorrect operation.

(10) The at least one operating region is preferably an essentially flat sub-surface of an outer surface of the household appliance. For example, a household appliance can have one operating region on its upper face and/or its front face respectively. The operating region is also preferably integrated in the respective outer surface of the household appliance so there are no intrusive transitions and/or edges between the outer surface and the operating region. The operating region is designed to control the household appliance. To this end, the operating region has input and output elements for example. These are, for example, buttons, switches, rotary controls, slide controls and/or touch-sensitive input elements. The operating region also has output elements, which include, for example, an LED (light emitting diode), segment display and/or LCD (liquid crystal display). The output elements are designed, in particular, to display a current operating state of the household appliance. The current operating state includes for example information relating to functions currently in use, set temperatures and/or power values, a set cooking time and/or further parameters relating to the household appliance. The household appliance also has an electronic control system (not described in detail), which is designed, in particular, to process inputs by way of the operating region, to activate the operating region to output operating parameters and to activate the functional units of the household appliance, such as a hotplate and/or heating coil for example.

(11) The operating element is designed, in particular, to operate the operating region. It has a two-part structure comprising the inner element and the outer element. The inner element is configured as an operating element base, for example, and the outer element is configured as an operating element top, the two elements being connected by way of a ball bearing for example, and therefore, being able to be rotated relative to one another. The operating element base preferably has a flat lower face, which can be positioned on the operating region. The flat surface means there is a large contact surface with the operating region, ensuring that the operating element sits securely on the operating region. The operating element can be used, in particular, as a rotary control when it is arranged on the operating region. It is therefore possible to dispense with other rotary controls arranged permanently on the operating region, thereby enhancing the user-friendliness of the household appliance.

(12) A retaining force for the operating element, which holds the operating element securely on the operating region, for example when the operating region is arranged vertically on the front face of the household appliance, is provided by a magnetic force between the operating element and the operating region. To this end the operating region comprises for example a magnetizable material, so that a magnetic retaining force is established between the magnetic body of the operating element and the magnetizable material. Magnetic bodies can also be provided in the operating region, for example to increase the magnetic retaining force and/or to predefine a magnetic field with a defined direction.

(13) The magnetic body of the operating element is, in particular, a permanent magnet with a magnetic field, which is powerful enough to hold the operating element securely on the operating region. A magnet has a north pole and a south pole for example. Magnetic field lines form between the north pole and the south pole in permanently closed loops. The field lines, for example, run from the south pole to the north pole in a magnetic body and from the north pole to the south pole outside the magnetic body. A magnetic field can be described formally as a vector field, having a defined size and direction at every position. The field lines of a magnetic field show the local direction of the magnetic field, while the strength of the magnetic field is defined by the number of field lines running through a defined surface. If the magnetic fields of two magnetic bodies are overlaid, a magnetic force acts on the two magnetic bodies such that each tries to align itself along the field lines of the other magnetic body. It can also be said that a magnet with a magnetic field moved into a predefined magnetic field has a potential (magnetic) energy that it tries to minimize

by alignment along the field lines of the predefined magnetic field. This alignment force is also referred to here as a magnetic orientation force. For this reason, the magnetic body of the operating element will also try to align itself correspondingly in a predefined magnetic field and with it the inner element of the operating element. Such alignment of the inner element takes place, in particular, relative to the outer element, which is held by a user for example.

(14) The inner element of the operating element also has a user interface element. The user interface element is designed for example to display operating parameters of the household appliance and/or to receive user inputs for controlling the household appliance. The user interface element can comprise a touch-sensitive display for example. Because the inner element and with it the user interface element align themselves according to the predefined magnetic field, it is ensured that a graphic display on the user interface element for example can always be easily read by a user.

(15) According to one embodiment of the household appliance, the at least one operating region and the user interface element of the operating element can be configured as a function of a position of the operating element relative to the operating region.

(16) This embodiment has the advantage, in particular, that a display on the operating region and/or on the user interface element can be selected as a function of position and context. This ensures for example that a display element of the operating region is not covered by the operating element. A number of positions can also be provided on the operating region for example, with the operating element functioning as a rotary knob in a first position, the operating element functioning as a clock display in a second position and the operating element functioning as a selector switch to activate different functions of the household appliance in a third position.

(17) According to a further embodiment of the household appliance the magnetic body of the operating element is configured as a ring magnet.

(18) The ring magnet, in particular, has a diametrical arrangement of its two poles. The cylindrically symmetrical mass distribution of such a ring magnet means that it is particularly suitable for the operating element, for example a rotary knob. Such a ring magnet can also be oriented simply and reliably by means of a corresponding ring magnet.

(19) According to a further embodiment of the household appliance the magnetic body of the operating element is configured as a bar magnet or a curved bar magnet.

(20) A bar magnet can also be referred to as a magnet block.

(21) A curved bar magnet is a bar magnet which does not run precisely straight but along a circle radius for example. Such a curved bar magnet does not have a contour with point symmetry, in particular. A curved bar magnet can also be referred to as a magnet segment.

(22) According to a further embodiment of the household appliance, a number of magnetic bodies are provided in the operating element.

(23) Provision can be made for example for two bar magnets to be arranged in the operating element. This allows complex incorporation situations to be achieved for example and/or a magnetic retaining and/or orientation force to be increased.

(24) According to a further embodiment of the household appliance the at least one operating region has at least one predefining apparatus for predefining the magnetic field.

(25) The predefining apparatus comprises for example permanently magnetic materials and/or at least one electromagnet. The operating region preferably has a number of predefining apparatuses. In particular, the operating region can have a specific predefining apparatus in different preferred positions for the operating element.

(26) According to a further embodiment of the household appliance the at least one predefining apparatus for predefining the magnetic field is configured as a magnetic body corresponding to the magnetic body of the operating element.

(27) A corresponding magnetic body here means that it has for example the same size and/or geometry as the magnetic body of the inner element of the operating element. If the magnetic body of the inner element of the operating element is configured as a bar magnet for example, the

magnetic body of the predefining apparatus is preferably also configured as a bar magnet. Such corresponding configuration of the two magnetic bodies ensures a powerful magnetic retaining force and a powerful magnetic orientation force and therefore reliable alignment of the inner element of the operating element in the predefined magnetic field. It also defines a retaining position of the operating element on the operating region exactly, as the magnetic force between the two magnetic bodies also brings about exact positioning of the magnets relative to one another.

(28) According to a further embodiment of the household appliance, the magnetic body of the operating element is configured as a diametrically magnetized ring magnet and the at least one predefining apparatus is configured as a diametrically magnetized cylindrical magnet.

(29) This embodiment results in a particularly powerful magnetic retaining force with a powerful magnetic orientation force.

(30) According to a further embodiment of the household appliance, the at least one predefining apparatus and the magnetic body of the operating element are matched to one another such that alignment of the inner element of the operating element by the predefined magnetic field takes place when the operating element is at a distance of 1 to 20 mm, preferably 5 to 15 mm, from the operating region.

(31) This embodiment advantageously ensures that the inner element of the operating element is already aligned according to the predefined magnetic field when the operating element is positioned on the operating region, thereby ensuring a powerful magnetic retaining force. The operating element is practically prevented from falling down. It also allows exact positioning of the operating element, as the magnetic force pulls the operating element into a position exactly above the predefining apparatus, if the operating element is positioned for example slightly out of place.

(32) According to a further embodiment of the household appliance the outer element of the operating element can be rotated relative to the inner element of the operating element when the operating element is positioned on the operating region.

(33) In this embodiment the operating element can be used, in particular, as a rotary knob.

(34) According to a further embodiment of the household appliance the user interface element has a display, which is designed to display at least one operating parameter of the household appliance graphically.

(35) The automatic alignment of the inner element of the operating element when the operating element is positioned on the operating region advantageously ensures that the graphic display always has the correct orientation. This allows complex information, such as numbers, letters and/or words, also to be displayed on the display. In this embodiment it is advantageously possible to dispense with further graphic display elements, which are integrated for example in one of the operating regions. This reduces the complexity of the household appliance and can, in particular, help to reduce costs.

(36) According to a further embodiment of the household appliance the display is configured as a touch-sensitive display.

(37) As well as the advantages described above, this embodiment also has the further advantage that complex user inputs are also possible. For example, the touch-sensitive display can be designed to display one or more switches.

(38) According to a further embodiment of the household appliance the inner element of the operating element has an energy supply apparatus to supply the user interface element with electrical energy.

(39) The energy supply apparatus can comprise a battery, an accumulator and/or a charging facility for example. A charging facility includes different embodiments, such as contact-based embodiments and contactless embodiments.

(40) According to a further embodiment of the household appliance the energy supply apparatus has two contact points, the operating region also having two corresponding contact points and the contact points being in electrical contact with the corresponding contact points when the operating

element is positioned on the operating region.

(41) According to a further embodiment of the household appliance the energy supply apparatus has a charging coil, the operating region also having at least one transmit coil, by means of which electrical energy can be transferred inductively to the charging coil.

(42) In this embodiment the energy supply facility is configured contactlessly. This has the advantage, in particular, that there are no open contacts arranged on the household appliance and/or on the operating element, which could potentially represent a safety hazard. It also allows continuous operation without a user having to replace a battery for example. As the magnetic body of the inner element ensures alignment along the predefined magnetic field, the efficiency of the transfer by means of the inductive energy transfer is also advantageously ensured.

(43) Provision can be made for example for the operating element to have an accumulator, which supplies the user interface element with electrical energy, when the operating element is used. A charging position for the operating element can also be provided, to which the operating element is moved when it is not in use and in which the accumulator is charged.

(44) The household appliance is configured, in particular, as a cooking appliance, a cooktop, in particular an induction cooktop, an oven, a microwave, a grill and/or a steam cooking appliance.

(45) According to a further aspect an operating element for operating a household appliance, in particular a household appliance according to one of the embodiments described above, is proposed. The operating element has an inner and outer element, the inner element being able to be rotated relative to the outer element and the inner element having a user interface element and a magnetic body. The inner element can be aligned relative to the outer element by the magnetic body by means of a predefined magnetic field.

(46) Such an operating element has all the advantages described in relation to the embodiments of the household appliance described above.

(47) According to one embodiment of the operating element it is configured as a removable rotary knob.

(48) Such an embodiment has the advantage that a haptic operating experience is possible for users of the household appliance with this operating element.

(49) According to a further aspect a method for controlling a household appliance with an operating region is proposed. The method comprises the following steps in particular: bringing an operating element toward the operating region, it being possible for the operating element to be held on the operating region by means of magnetic force and to be moved relative to the operating region and to be positioned in such a manner that it can be removed therefrom. The operating element has an inner element and an outer element, the inner element being able to be rotated relative to the outer element. The inner element has a user interface element and a magnetic body, by which the inner element can be aligned relative to the outer element by means of a predefined magnetic field; aligning the inner element of the operating element according to the predefined magnetic field; and positioning the operating element on the operating region to operate the household appliance. Operating the household appliance with the operating element, in particular, includes displaying an operating parameter of the household appliance and/or acquiring a user input to control the household appliance.

(50) The embodiments and features described for the proposed apparatus apply correspondingly to the proposed method.

(51) Further possible implementations of the invention comprise combinations of features or embodiments described above or in the following with regard to the exemplary embodiments even if these are not cited specifically. The person skilled in the art will also add individual aspects to improve or complete the respective basic form of the invention.

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Further advantageous configurations and aspects of the invention are set out in the subclaims and the exemplary embodiments of the invention described in the following. The invention is also described in more detail based on preferred embodiments with reference to the accompanying figures.
- (2) FIG. 1 shows a schematic view of an exemplary embodiment of a household appliance with an operating region and an operating element;
- (3) FIG. 2a shows a schematic plan view of an exemplary embodiment of an operating element with a ring magnet;
- (4) FIG. 2b shows a schematic section through a ring magnet with diametrically arranged poles and its magnetic field;
- (5) FIG. 3 shows a schematic section through an embodiment of the household appliance with a vertically arranged operating region and an operating element arranged thereon;
- (6) FIGS. 4a to 4c show a schematic view of an exemplary embodiment of a household appliance configured as a cooktop with an operating region and an operating element; and
- (7) FIG. 5 shows an exemplary embodiment of a method for controlling a household appliance.

DETAILED DESCRIPTION OF EXEMPLARY EMBODIMENTS OF THE PRESENT INVENTION

- (8) Identical elements or those of identical function are shown with the same reference characters in the figures, unless otherwise specified.
- (9) FIG. 1 shows a schematic view of a first exemplary embodiment of a household appliance **10**. The household appliance **10** is configured here as a cuboid cooker comprising an oven (not shown) and a cooktop **11**, which comprises four cooking zones **12a-12d**. The household appliance **10** has a vertically arranged operating region **20** on its front face. An operating element **30** is shown next to the household appliance **10**. The operating element **30** comprises an inner element **31** and an outer element **32**. The inner element **31** has a user interface element **33** and a magnetic body **34**. The operating element **30** here is configured as a round operating knob **30**. The user interface element **33** is configured by way of example as an LCD and the magnetic body **34** is configured by way of example as a bar magnet.
- (10) The operating element **30** is positioned on the operating region **20** to operate the household appliance **10**. Provision can be made for example for the operating element **30** to be positioned in the center of the operating region **20** to control the cooktop **11**. A cooking zone **12a-12d** of the cooktop **11** can now be selected first. To this end numbers from 1 to 4 are displayed on the user interface element **33** configured as an LCD of the operating element **30** for example, it being possible to switch between the numbers and therefore between the cooking zones **12a-12d** by rotating the outer element **32**. One of the cooking zones **12a-12d** is selected by pressing the outer element **32** for example. A cooking stage is then selected for example by rotating the outer element **32**. A circular segment display for example is displayed on the operating region **20** to show the cooking stage, being displayed as a function of a degree of rotation of the outer element **32**. Alternatively or additionally, the set cooking stage can be displayed as a number from 0-9 or even from 0-100 on the user interface element **33** configured as an LCD of the operating element **30**.
- (11) At the end of the cooking operation the operating element **30** can be removed from the operating region **20**. This moves the household appliance **10** (e.g., cooker) into a rest mode, in which in particular all the cooking zones **12a-12d** of the cooktop **11** are deactivated. The operating element **30** can then be moved for example into a standby position (not shown in detail), in which the user interface element **33** configured as an LCD can be used for example to display the current time.
- (12) FIG. 2a shows a detailed view of an exemplary embodiment of the operating element **30**. The operating element **30** comprises an inner element **31** and an outer element **32**. The inner element **31**

has a user interface element **33** (not shown here, see for example FIG. **1**) and also a magnetic body **34** configured as a ring magnet. A preferred magnetic direction **B** of the magnetic body **34** configured as a ring magnet is shown by means of an arrow in FIG. **2a**. The operating element **30** can have a different shape from the one in FIG. **2a**. It can be configured as a block or a different, in particular regular, polygon for example.

(13) FIG. **2b** shows a section through the magnetic body **34** configured as a ring magnet from FIG. **2a**. The two magnetic poles **N**, **S** of the magnetic body **34** configured as a ring magnet are shown in particular and the magnetic field of the magnetic body **34** configured as a ring magnet is indicated by field lines **40**. The field lines **40** have only been shown symbolically in the upper half plane here but the field pattern in the lower half plane can also be obtained by reflection along the plane of symmetry **SE**. The preferred direction **B** of the magnetic body **34** configured as a ring magnet is also shown.

(14) FIG. **3** shows a schematic view of a section through a further exemplary embodiment of a household appliance **10** with a vertically arranged operating region **20**. The household appliance **10** in FIG. **3** is for example the household appliance **10** (e.g., cooker) from FIG. **1**. The operating region **20** has two predefining apparatuses **M1**, **M2** configured as magnetic bodies. Each magnetic body **M1**, **M2** has its own magnetic field **B1**, **B2**, it being possible for the preferred direction of the different magnetic fields **B1**, **B2** to be different. Each predefining apparatus **M1**, **M2** is also assigned a transmit coil **21**, **22** for transferring energy inductively to a charging coil **35**. In the example in FIG. **3** the operating element **30** is arranged on the operating region **20** on the magnetic body **M1**. The operating element **30** has an inner element **31** and an outer element **32**. The inner element **31** also has a user interface element **33** configured as an LCD, a magnetic body **34** with a preferred direction **B** and a charging coil **35**.

(15) The front face of the outer element **32**, facing left in FIG. **3**, is made for example of an at least partially transparent material such as glass or plastic, to allow the user interface elements **33** configured as LCDs to be read. In the example, the magnetic body **34** has aligned itself according to the magnetic field **B1** predefined by the corresponding magnetic body **M1**. The alignment of the inner element **31** is therefore generally known, ensuring that a graphic display on the user interface element **33** configured as an LCD is correctly aligned and can therefore be read by a user. The charging coil **35** is designed to receive electrical energy inductively from the transmit coil **21** of the operating region **20**, allowing the user interface element **33** configured as an LCD for example to be operated.

(16) FIG. **4a** shows a schematic view of a further exemplary embodiment of a household appliance **10**, configured here as an induction cooktop **11**. The induction cooktop **11** has two cooking zones **12a**, **12b** and an operating region **20**, to which three predefining apparatuses **M1**, **M2**, **M3** are assigned. The predefining apparatuses **M1**, **M2**, **M3** are configured as ring magnets here, each of the ring magnets **M1**, **M2**, **M3** having its own magnetic field **B1**, **B2**, **B3**. The regions on the ring magnets **M1**, **M2**, **M3** in this example are preferred positions **P1**, **P2**, **P3** for the operating element **30**. The operating element **30** is shown away from the induction cooktop **11** in FIG. **4a**. It has an inner element **31** and an outer element **32**. The inner element **31** has a user interface element **33** and a magnetic body **34** configured as a ring magnet, which has not been shown in FIG. **4a** for reasons of clarity (see for example FIGS. **2a** and **2b**). The alignment of the inner element **31** is indicated by cross-hatching. In the initial position in FIG. **4a** the inner element **31** is aligned at an arbitrary angle to the induction cooktop **11**.

(17) If a user now wishes to activate the right cooking zone **12b** of the induction cooktop **11**, said user positions the operating element **30** for example in position **P3** on the operating region **20**, as shown in FIG. **4b**. As shown by the hatching of the inner element **31** of the operating element **30**, the inner element **31** has aligned itself according to the magnetic field **B3** predefined by the ring magnet **M3** as it is brought toward the position **P3**. This ensures that an operating parameter is displayed on the user interface element **33** correctly and in such a manner that it can be read

reliably by the user. The reference characters for the operating element **30** and the features associated with the position **P3** are not shown in FIG. **4b** in order not to overload the diagram.

(18) Once the user has for example set a cooking stage for the right cooking zone **12b** of the induction cooktop **11** by means of the operating element **30**, said user may wish to input a cooking time for example. To this end, a provision can be made for the user to remove the operating element from the position **P3** and reposition it in the position **P2**. The magnetic body **M2** characterizing the position **P2** has a magnetic field **B2**, which is aligned perpendicular to the magnetic field **B3** of the magnet **M3** in the example. Therefore, the alignment of the inner element **31** of the operating element **30** also changes, when the operating element **30** is brought toward the position **P2**, as shown by the hatching of the inner element **31**. The reference characters for the operating element **30** and the features associated with the position **P2** are not shown in FIG. **4c** in order not to overload the diagram.

(19) FIG. **5** shows an exemplary embodiment of a method for controlling a household appliance **10** with an operating region **20**, for example the household appliance **10** in FIG. **1** or the induction cooktop **11** in FIGS. **4a-4c**.

(20) In step **S1** of the method, an operating element **30** is brought toward the operating region **20** of the household appliance **10**. The operating region **20** has a magnetic body **M1** with a magnetic field **B1**. The operating element **30** has a two-part structure with an inner element **31** and an outer element **32**. The inner element **31** has a user interface element **33** and a magnetic body **34**.

(21) In step **S2** of the method the inner element **31** of the operating element **30** is aligned according to the predefined magnetic field **B1**. This takes place in particular when the operating element **30** is at a distance of around 20 mm to 1 mm from the operating region **20**.

(22) In step **S3** the operating element **30** is positioned on the operating region **20** and can then be used to operate the household appliance **10**. Operate here means that for example an operating parameter of the household appliance **10** is displayed to a user by means of the user interface element **33** of the operating element **30** and/or a user input is acquired to control the household appliance **10**. The household appliance **10** is first activated for example by pressing the outer element **32** of the operating element **30**. A function of the household appliance **10**, for example an oven or cooking zone **12a-12d**, is then selected by rotating the outer element **32**. Selection can take place for example by showing a graphic display on the operating region **20** and/or on the user interface element **33**. The currently selected function can be activated for example by pressing the outer element **32** again. A current cooking chamber temperature of the oven for example can then be displayed on the user interface element **33**. To terminate the cooking operation, the operating element **30** is for example simply removed from the operating region **20** and moved to a rest position.

(23) Although the present invention has been described based on exemplary embodiments, it can be modified in many different ways.

Claims

1. A household appliance, comprising: an operating region for controlling the household appliance; and an operating element configured for attachment on the operating region by magnetic force and removal therefrom and being movable relative to the operating region, said operating element including an outer element and an inner element which is rotatable relative to the outer element, said inner element including a user interface element and a magnetic body, by which the inner element is capable of being aligned relative to the outer element by a predefined magnetic field, wherein the operating element is secured on the operating region by a magnetic retaining force, wherein the predefined magnetic field of the magnetic body has a defined direction, and wherein the inner element of the operating element is capable of rotating relative to the outer element and is capable of being rotationally aligned with respect to a particular control location of the operating

region by the predefined magnetic field having the defined direction when the operating element is being secured on the operating region, thereby aligning the user interface element of the inner element with the particular control location of the operating region.

2. The household appliance of claim 1, wherein the operating region and the user interface element of the operating element are configured as a function of a position of the operating element relative to the operating region.
3. The household appliance of claim 1, wherein the magnetic body of the operating element is configured as a ring magnet.
4. The household appliance of claim 1, wherein the magnetic body of the operating element is configured as a bar magnet or a curved bar magnet.
5. The household appliance of claim 1, further comprising a predefining apparatus provided in the operating region for predefining the magnetic field.
6. The household appliance of claim 5, wherein the predefining apparatus is configured as a magnetic body corresponding to the magnetic body of the operating element.
7. The household appliance of claim 5, wherein the predefining apparatus and the magnetic body of the operating element are matched such that alignment of the inner element of the operating element by the predefined magnetic field takes place when the operating element is at a distance of 1-20 mm from the operating region.
8. The household appliance of claim 7, wherein the distance is 5-15 mm.
9. The household appliance of claim 1, wherein the outer element of the operating element is rotatable relative to the inner element of the operating element, when the operating element is positioned on the operating region.
10. The household appliance of claim 1, wherein the user interface element includes a display configured to display an operating parameter of the household appliance graphically.
11. The household appliance of claim 10, wherein the display is configured as a touch-sensitive display.
12. The household appliance of claim 1, wherein the inner element of the operating element includes an energy supply apparatus to supply the user interface element with electrical energy.
13. The household appliance of claim 12, wherein the energy supply apparatus includes a charging coil, said operating region including a transmit coil for transfer of electrical energy inductively to the charging coil.
14. The household appliance of claim 1, wherein the inner element is radially inward of the outer element.
15. An operating element for operating a household appliance, comprising: an outer element; and an inner element, which is radially inward of the outer element and rotatable relative to the outer element, said inner element including a user interface element and a magnetic body, by which the inner element is capable of being aligned relative to the outer element by a predefined magnetic field, wherein the operating element is configured to be secured on an operating region of the household appliance by a magnetic retaining force, wherein the predefined magnetic field of the magnetic body has a defined direction, and wherein the inner element of the operating element is configured to rotate relative to the outer element to rotationally align with respect to a particular control location of the operating region by the predefined magnetic field having the defined direction when the operating element is being secured on the operating region to thereby align the user interface element of the inner element with the particular control location of the operating region.
16. The operating element of claim 15, wherein the operating element is embodied as a removable rotary knob.
17. The operating element of claim 15, wherein the user interface element is configured as a function of a position of the operating element relative to the operating region of the household appliance.

18. The operating element of claim 15, wherein the magnetic body is configured as a magnet selected from the group consisting of a ring magnet, a bar magnet, and a curved bar magnet.
19. The operating element of claim 15, wherein the outer element is configured to be rotatable relative to the inner element of the operating element when the operating element is positioned on the operating region of the household appliance.
20. The operating element of claim 15, wherein the user interface element includes a display configured to display an operating parameter of the household appliance graphically.
21. The operating element of claim 20, wherein the display is configured as a touch-sensitive display.
22. The operating element of claim 15, wherein the inner element includes an energy supply apparatus to supply the user interface element with electrical energy.
23. The operating element of claim 22, wherein the energy supply apparatus includes a charging coil for receiving electrical energy inductively from a transmit coil of an operating region of the household appliance.
24. A method for controlling the household appliance of claim 1, said method comprising: bringing the operating element toward the operating region to enable the operating element to be held on the operating region by magnetic force and to be moved relative to the operating region and to be positioned for removal therefrom; aligning the inner element of the operating element according to the predefined magnetic field; and positioning the operating element on the operating region to operate the household appliance, thereby at least one of enabling display of an operating parameter of the household appliance and acquiring a user input to control the household appliance.
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